

IIT MATHEMATICS

METHODS OF DIFFERENTIATION

WORKBOOK



RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
Gen. Category - 2014



AIR-3
GOVIND LAHOTI
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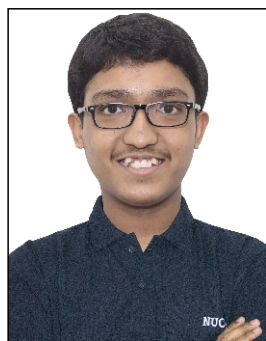


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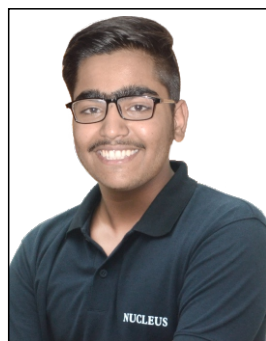


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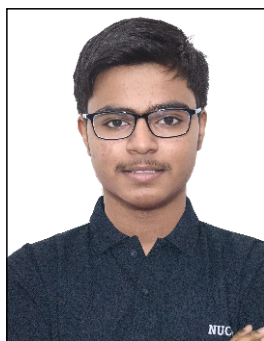
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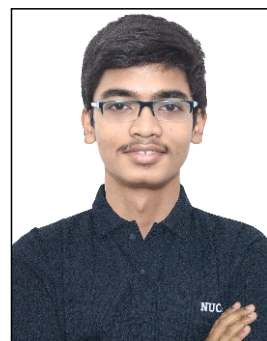
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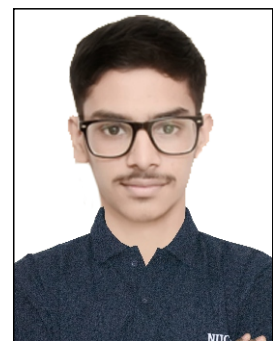
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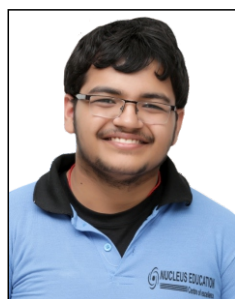


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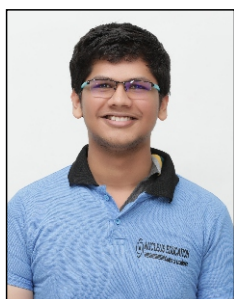


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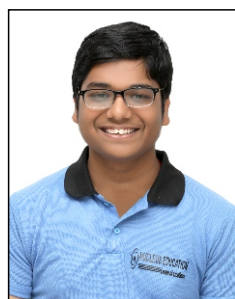
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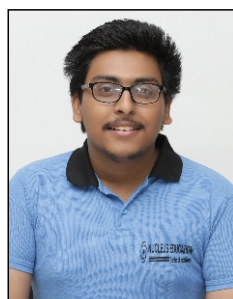
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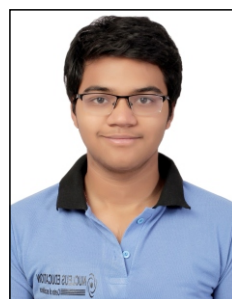
AIR-12
YATEESH AGRAWAL
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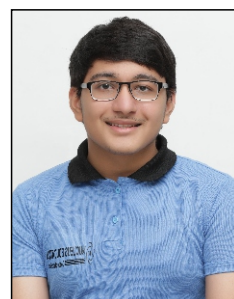
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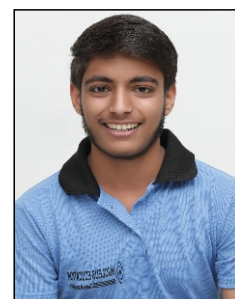
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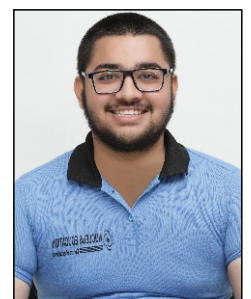
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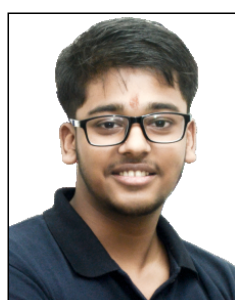
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MAYANK DUBEY
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SHASHANK AGRAWAL
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RAAGHAV RAAJ
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AIR-61
SARTHAK BEHERA
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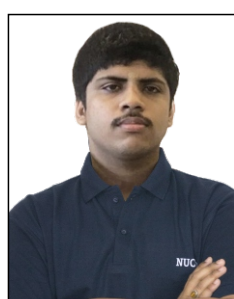
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HIMANSHU GAURAV SINGH
Gen. Category - 2019
*SDCCP



AIR-19
VIBHAV AGGARWAL
Gen. Category - 2019



AIR-33
S. PRAJEETH
Gen. Category - 2019
*SDCCP



AIR-48
SOHAM MISTRI
Gen. Category - 2019
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AIR-51
SAYANTAN DHAR
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METHODS OF DIFFERENTIATION

The process of calculating derivative is called differentiation.

1. DERIVATIVE OF $f(x)$ FROM THE FIRST PRINCIPLE :

Obtaining the derivative using the definition $\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0}$

$\frac{f(x+\delta x) - f(x)}{\delta x} = f'(x) = \frac{dy}{dx}$ is called calculating derivative using first principle or ab initio or delta method.

Note : $\frac{dy}{dx}$ can also be represented as y_1

or y' or Dy or $f'(x)$. $\frac{dy}{dx}$ represents instantaneous rate of change of y w.r.t. x .

Illustration 1 :

Differentiate each of following functions by first principle :

(i) $f(x) = \tan x$

(ii) $f(x) = e^{\sin x}$

Solution :

$$\begin{aligned} \text{(i)} \quad f'(x) &= \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\tan(x+h-x)[1 + \tan x \tan(x+h)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{\tanh}{h} \cdot (1 + \tan^2 x) = \sec^2 x. \end{aligned}$$

Ans.

$$\begin{aligned} \text{(ii)} \quad f'(x) &= \lim_{h \rightarrow 0} \frac{e^{\sin(x+h)} - e^{\sin x}}{h} \\ &= \lim_{h \rightarrow 0} e^{\sin x} \frac{[e^{\sin(x+h) - \sin x} - 1]}{\sin(x+h) - \sin x} \left(\frac{\sin(x+h) - \sin x}{h} \right) \\ &= e^{\sin x} \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = e^{\sin x} \cos x \end{aligned}$$

Ans.

Do yourself -1 :

(i) Differentiate each of following functions by first principle:

(a) $f(x) = \ln x$

(b) $f(x) = \frac{1}{x}$

2. DERIVATIVE OF STANDARD FUNCTIONS :

	$f(x)$	$f'(x)$		$f(x)$	$f'(x)$
(i)	x^n	nx^{n-1}	(ii)	e^x	e^x
(iii)	a^x	$a^x \ln a, a > 0$	(iv)	$\ln x$	$1/x$
(v)	$\log_a x$	$(1/x) \log_a e, a > 0, a \neq 1$	(vi)	$\sin x$	$\cos x$
(vii)	$\cos x$	$-\sin x$	(viii)	$\tan x$	$\sec^2 x$
(ix)	$\sec x$	$\sec x \tan x$	(x)	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cdot \cot x$
(xi)	$\cot x$	$-\operatorname{cosec}^2 x$	(xii)	constant	0
(xiii)	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$	(xiv)	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$

(xv)	$\tan^{-1} x$	$\frac{1}{1+x^2}, x \in \mathbb{R}$	(xvi)	$\sec^{-1} x$	$\frac{1}{ x \sqrt{x^2-1}}, x > 1$
(xvii)	$\operatorname{cosec}^{-1} x$	$\frac{-1}{ x \sqrt{x^2-1}}, x > 1$	(xviii)	$\cot^{-1} x$	$\frac{-1}{1+x^2}, x \in \mathbb{R}$

3. FUNDAMENTAL THEOREMS :

If f and g are derivable functions of x , then,

- (a) $\frac{d}{dx}(f \pm g) = \frac{df}{dx} \pm \frac{dg}{dx}$
 (b) $\frac{d}{dx}(cf) = c \frac{df}{dx}$, where c is any constant
 (c) $\frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx}$ known as

"PRODUCT RULE"

- (d) $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\left(\frac{df}{dx}\right) - f\left(\frac{dg}{dx}\right)}{g^2}$ where $g \neq 0$ known as "QUOTIENT RULE"

- (e) If $y = f(u)$ & $u = g(x)$ then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ known as "CHAIN RULE"

Note : In general if $y = f(u)$ then

$$\frac{dy}{dx} = f'(u) \cdot \frac{du}{dx}.$$

Illustration 2 :

If $y = e^x \tan x + x \log_e x$, find $\frac{dy}{dx}$.

Solution :

$y = e^x \cdot \tan x + x \cdot \log_e x$
 On differentiating we get,

$$\frac{dy}{dx} = e^x \cdot \tan x + e^x \cdot \sec^2 x + 1 \cdot \log x + x \cdot \frac{1}{x}$$

$$\text{Hence, } \frac{dy}{dx} = e^x(\tan x + \sec^2 x) + (\log x + 1)$$

Ans.

Illustration 3 :

If $y = \frac{\log x}{x} + e^x \sin 2x + \log_5 x$, find $\frac{dy}{dx}$.

Solution :

On differentiating we get,

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\log x}{x} \right) + \frac{d}{dx} (e^x \sin 2x) + \frac{d}{dx} (\log_5 x)$$

$$= \frac{1}{x} \cdot x - \log x \cdot 1 + \frac{d}{dx} (e^x \sin 2x) + \frac{1}{x \log_e 5}$$

$$\text{Hence, } \frac{dy}{dx} = \left(\frac{1 - \log x}{x^2} \right)$$

$$+ e^x(\sin 2x + 2 \cos 2x) + \frac{1}{x \log_e 5}$$

Ans.

Illustration 4 :

If $y = \log_e(\tan^{-1} \sqrt{1+x^2})$, find $\frac{dy}{dx}$.

Solution :

$$y = \log_e (\tan^{-1} \sqrt{1+x^2})$$

On differentiating we get,

$$= \frac{1}{\tan^{-1} \sqrt{1+x^2}} \cdot \frac{1}{1+(\sqrt{1+x^2})^2} \cdot \frac{1}{2\sqrt{1+x^2}} \cdot 2x$$

$$= \frac{x}{\left(\tan^{-1} \sqrt{1+x^2}\right) \left\{1+(\sqrt{1+x^2})^2\right\} \sqrt{1+x^2}}$$

$$= \frac{x}{\left(\tan^{-1} \sqrt{1+x^2}\right) (2+x^2) \sqrt{1+x^2}} \quad \text{Ans.}$$

Do yourself -2 :

(i) Find $\frac{dy}{dx}$ if -

- (a) $y = (x+1)(x+2)(x+3)$
 (b) $y = e^{5x} \tan(x^2+2)$

4. LOGARITHMIC DIFFERENTIATION :

To find the derivative of a function :

- (a) which is the product or quotient of a number of functions or
 (b) of the form $[f(x)]^{g(x)}$ where f & g are both derivable functions.
 It is convenient to take the logarithm of the function first & then differentiate.

Illustration 5 :

If $y = (\sin x)^{nx}$, find $\frac{dy}{dx}$

Solution :

$$\ln y = \ln x \cdot \ln (\sin x)$$

On differentiating we get,

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} \ln (\sin x) + \ln x \cdot \frac{\cos x}{\sin x}$$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^{nx} \left[\frac{\ln(\sin x)}{x} + \cot x \ln x \right]$$

Ans.

Illustration 6 :

If $x = \exp \left(\tan^{-1} \left(\frac{y-x^2}{x^2} \right) \right)$, then $\frac{dy}{dx}$ equals -

- (A) $x [1 + \tan (\log x) + \sec^2 x]$
 (B) $2x [1 + \tan (\log x)] + \sec^2 x$
 (C) $2x [1 + \tan (\log x)] + \sec x$
 (D) $2x + x[1 + \tan(\log x)]^2$

Solution :

Taking log on both sides, we get

$$\log x = \tan^{-1} \left(\frac{y-x^2}{x^2} \right)$$

$$\Rightarrow \tan (\log x) = (y-x^2) / x^2$$

$$\Rightarrow y = x^2 + x^2 \tan (\log x)$$

On differentiating, we get

$$\therefore \frac{dy}{dx} = 2x + 2x \tan (\log x) + x \sec^2 (\log x)$$

$$\Rightarrow 2x [1 + \tan (\log x)] + x \sec^2 (\log x) = 2x + x[1 + \tan(\log x)]^2$$

Ans. (D)

Illustration 7 :

If $y = \frac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$ find $\frac{dy}{dx}$

Solution :

$$\ln y = \frac{1}{2} \ln x + \frac{2}{3} \ln (1-2x) - \frac{3}{4} \ln (2-3x) - \frac{4}{5} \ln (3-4x)$$

On differentiating we get,

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)}$$

$$\frac{dy}{dx} = y \left(\frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)} \right)$$

Ans.

Do yourself -3 :

(i) Find $\frac{dy}{dx}$ if $y = x^x$

(ii) Find $\frac{dy}{dx}$ if $y = e^x \cdot e^{x^2} \cdot e^{x^3} \cdot e^{x^4}$

5. PARAMETRIC DIFFERENTIATION :

If $y = f(\theta)$ & $x = g(\theta)$ where θ is a parameter, then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{f'(\theta)}{g'(\theta)}$

Illustration 8 :

If $y = a \cos t$ and $x = a(t - \sin t)$ find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{2}$

Solution :

$$\frac{dy}{dx} = \frac{-a \sin t}{a(1 - \cos t)} \Rightarrow \frac{dy}{dx} \Big|_{t=\frac{\pi}{2}} = -1$$

Ans.

Illustration 9 :

Prove that the function represented parametrically by the equations.

$x = \frac{1+t}{t^3}$; $y = \frac{3}{2t^2} + \frac{2}{t}$ satisfies the relationship : $x(y')^3 = 1 + y'$ (where $y' = \frac{dy}{dx}$)

Solution :

$$\text{Here } x = \frac{1+t}{t^3} = \frac{1}{t^3} + \frac{1}{t^2}$$

Differentiating w.r. to t

$$\frac{dx}{dt} = -\frac{3}{t^4} - \frac{2}{t^3}$$

$$y = \frac{3}{2t^2} + \frac{2}{t}$$

Differentiating w.r. to t

$$\frac{dy}{dt} = -\frac{3}{t^3} - \frac{2}{t^2}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = t = y'$$

$$\text{Since } x = \frac{1+t}{t^3}$$

$$\Rightarrow x = \frac{1+y'}{(y')^3} \text{ or } x(y')^3 = 1 + y'$$

Ans.

6. DERIVATIVE OF A FUNCTION W.R.T. ANOTHER FUNCTION :

Let $y = f(x)$; $z = g(x)$ then

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$$

Illustration 10 :

Differentiate $\log_e(\tan x)$ with respect to $\sin^{-1}(e^x)$.

Solution :

$$\begin{aligned} \frac{d(\log_e \tan x)}{d(\sin^{-1}(e^x))} &= \frac{\frac{d}{dx}(\log_e \tan x)}{\frac{d}{dx} \sin^{-1}(e^x)} \\ &= \frac{\cot x \cdot \sec^2 x}{e^x \cdot 1/\sqrt{1-e^{2x}}} = \frac{e^{-x} \sqrt{1-e^{2x}}}{\sin x \cos x} \end{aligned}$$

Ans.

Do yourself-4 :

- Find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ if $y = \cos^4 t$ & $x = \sin^4 t$.
- Find the slope of the tangent at a point $P(t)$ on the curve $x = at^2$, $y = 2at$.
- Differentiate x^{nx} with respect to $\ln x$.

7. DIFFERENTIATION OF IMPLICIT FUNCTIONS : $\phi(x, y) = 0$

- To find dy/dx of implicit functions, we differentiate each term w.r.t. x regarding y as a function of x & then collect terms with dy/dx together on one side.

- Also $\frac{dy}{dx} = -\frac{\frac{\partial \phi}{\partial x}}{\frac{\partial \phi}{\partial y}}$, where $\frac{\partial \phi}{\partial x}$ = partial

derivative of $\phi(x, y)$ w.r.t. x taking y as a constant and $\frac{\partial \phi}{\partial y}$ = partial derivative

- of $\phi(x, y)$ w.r.t. y taking x as a constant.
- In the case of implicit functions, generally, both x & y are present in answers of dy/dx .

Illustration 11 :

If $x^y + y^x = 2$, then find $\frac{dy}{dx}$.

Solution :

Let $u = x^y$ and $v = y^x$
 $u + v = 2 \Rightarrow \frac{du}{dx} + \frac{dv}{dx} = 0$

Now $u = x^y$ and $v = y^x$

$\Rightarrow \ln u = y \ln x$ and $\ln v = x \ln y$

$\Rightarrow \frac{1}{u} \frac{du}{dx} = \frac{y}{x} + \ln x \frac{dy}{dx}$ and $\frac{1}{v} \frac{dv}{dx} = \ln y + \frac{x}{y} \frac{dy}{dx}$

$= \ln y + \frac{x}{y} \frac{dy}{dx}$

$\Rightarrow \frac{du}{dx} = x^y \left(\frac{y}{x} + \ln x \frac{dy}{dx} \right)$ and $\frac{dv}{dx} = y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right)$

$= y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right)$

$\Rightarrow x^y \left(\frac{y}{x} + \ln x \frac{dy}{dx} \right) + y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) = 0$

$\Rightarrow \frac{dy}{dx} = - \frac{\left(y^x \ln y + x^y \cdot \frac{y}{x} \right)}{\left(x^y \ln x + y^x \cdot \frac{x}{y} \right)}$ **Ans.**

Aliter :

$\phi(x, y) = x^y + y^x - 2 = 0$

$\frac{dy}{dx} = - \frac{\partial \phi / \partial x}{\partial \phi / \partial y} = - \left(\frac{yx^{y-1} + y^x \ln y}{x^y \ln x + xy^{x-1}} \right)$

Illustration 12 :

If $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \cos x} \dots}}$, prove that

$\frac{dy}{dx} = \frac{(1+y) \cos x + y \sin x}{1 + 2y + \cos x - \sin x}$

Solution :

Given function is

$y = \frac{\sin x}{1 + \frac{\cos x}{1 + y}} = \frac{(1+y) \sin x}{1 + y + \cos x}$

or $y + y^2 + y \cos x = (1+y) \sin x$

.....(i)

Differentiate both sides with respect to x,

$\frac{dy}{dx} + 2y \frac{dy}{dx} + \frac{dy}{dx} \cos x - y \sin x$

$= (1+y) \cos x + \frac{dy}{dx} \sin x$

$\frac{dy}{dx} (1 + 2y + \cos x - \sin x)$

$= (1+y) \cos x + y \sin x$

or $\frac{dy}{dx} = \frac{(1+y) \cos x + y \sin x}{1 + 2y + \cos x - \sin x}$ **Ans.**

Aliter :

From (i) $\phi(x, y) = (1+y) \sin x - y - y^2 - y \cos x = 0$

$\frac{dy}{dx} = - \frac{\partial \phi / \partial x}{\partial \phi / \partial y} = - \frac{(1+y) \cos x + y \sin x}{\sin x - 1 - 2y - \cos x}$

$= \frac{(1+y) \cos x + y \sin x}{1 + 2y + \cos x - \sin x}$

Do yourself -5 :

- (i) Find $\frac{dy}{dx}$, if $x + y = \sin(x - y)$
- (ii) If $x^2 + xe^y + y = 0$, find y' , also find the value of y' at point $(0, 0)$.

8. DIFFERENTIATION BY TRIGONOMETRIC TRANSFORMATION :

Some Standard Substitutions :

Expression	Substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$ or $a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$ or $a \cot \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$ or $a \csc \theta$
$\sqrt{\frac{a+x}{a-x}}$ or $\sqrt{\frac{a-x}{a+x}}$	$x = a \cos \theta$ or $a \cos 2\theta$
$\sqrt{2ax - x^2}$	$x = a(1 - \cos \theta)$

Illustration 13 :

If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find

(i) $f'(2)$ (ii) $f'\left(\frac{1}{2}\right)$

(iii) $f'(1)$

Solution :

$x = \tan\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$\Rightarrow y = \sin^{-1}(\sin 2\theta)$

$$y = \begin{cases} \pi - 2\theta & \frac{\pi}{2} < 2\theta < \pi \\ 2\theta & -\frac{\pi}{2} \leq 2\theta \leq \frac{\pi}{2} \\ -(\pi + 2\theta) & -\pi < 2\theta < -\frac{\pi}{2} \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \pi - 2\tan^{-1}x & x > 1 \\ 2\tan^{-1}x & -1 \leq x \leq 1 \\ -(\pi + 2\tan^{-1}x) & x < -1 \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} -\frac{2}{1+x^2} & x > 1 \\ \frac{2}{1+x^2} & -1 < x < 1 \\ -\frac{2}{1+x^2} & x < -1 \end{cases}$$

(i) $f'(2) = -\frac{2}{5}$

(ii) $f'\left(\frac{1}{2}\right) = \frac{8}{5}$

(iii) $f'(1^+) = -1$ and $f'(1^-) = +1$
 $\Rightarrow f'(1)$ does not exist **Ans.**

Illustration 14 :

$$\frac{d}{dx} \left\{ \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right\} =$$

(A) $-\frac{1}{2}$ (B) 0

(C) $\frac{1}{2}$ (D) -1

Solution :

Let $y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right)$.

Put $x = \cos 2\theta$ $\theta \in \left(0, \frac{\pi}{2}\right]$

$\therefore y = \sin^2 \cot^{-1} \left(\sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}} \right)$

$= \sin^2 \cot^{-1}(\cot \theta)$

$\therefore y = \sin^2 \theta = \frac{1-\cos 2\theta}{2} = \frac{1-x}{2} = \frac{1}{2} - \frac{x}{2}$

$\therefore \frac{dy}{dx} = -\frac{1}{2}$. **Ans. (A)**

Illustration 15 :

Obtain differential coefficient of

$\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\cos^{-1} \frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}$

Solution :

Assume $u = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$,

$v = \cos^{-1} \frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}$

The function needs simplification before differentiation

Let $x = \tan\theta$; $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\therefore u = \tan^{-1} \left(\frac{\sec\theta-1}{\tan\theta} \right) = \tan^{-1} \left(\frac{1-\cos\theta}{\sin\theta} \right)$
 $= \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2}$

$v = \cos^{-1} \frac{1+\sec\theta}{2\sec\theta} = \cos^{-1} \frac{1+\cos\theta}{2}$
 $= \cos^{-1} \left(\cos \frac{\theta}{2} \right) = \frac{\theta}{2} \Rightarrow u = v$

$\therefore \frac{du}{dv} = 1$. **Ans.**

Do yourself : 6

- (i) If $y = \cos^{-1}(4x^3 - 3x)$, then find :
- (a) $f'\left(-\frac{\sqrt{3}}{2}\right)$, (b) $f'(0)$,
- (c) $f'\left(\frac{\sqrt{3}}{2}\right)$.

9. DERIVATIVE OF A FUNCTION AND ITS INVERSE FUNCTION :

If g is inverse of f , then

- (a) $g\{f(x)\} = x$
 $g'\{f(x)\}f'(x) = 1$
- (b) $f\{g(x)\} = x$
 $f'\{g(x)\}g'(x) = 1$

Illustration 16 :

If g is inverse of f and $f'(x) = \frac{1}{1+x^n}$,

then $g'(x)$ equals :-

- (A) $1+x^n$ (B) $1+[f(x)]^n$
 (C) $1+[g(x)]^n$ (D) none of these

Solution :

Since g is the inverse of f . Therefore
 $f(g(x)) = x$ for all x

$$\Rightarrow \frac{d}{dx} f(g(x)) = 1 \text{ for all } x$$

$$\Rightarrow f'(g(x)) g'(x) = 1$$

$$\Rightarrow g'(x) = \frac{1}{f'(g(x))} = 1 + (g(x))^n$$

Ans. (C)

Do yourself -7 :

- (i) If g is inverse of f and $f(x) = 2x + \sin x$;
 then $g'(x)$ equals:
- (A) $-\frac{3}{x^2} + \frac{1}{\sqrt{1-x^2}}$
 (B) $2 + \sin^{-1}x$
 (C) $2 + \cos(g(x))$
 (D) $\frac{1}{2 + \cos(g(x))}$

10. HIGHER ORDER DERIVATIVES :

Let a function $y = f(x)$ be defined on an interval (a, b) . If $f(x)$ is differentiable function, then its derivative $f'(x)$ [or (dy/dx) or y'] is called the first derivative of y w.r.t. x . If $f'(x)$ is again differentiable function on (a, b) , then its derivative $f''(x)$ [or d^2y/dx^2 or y''] is called second derivative of y w.r.t. x . Similarly, the 3rd order derivative of y w.r.t. x , if it exists, is defined by

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) \text{ and denoted by } f'''(x) \text{ or } y''' \text{ and so on.}$$

Note : If $x = f(\theta)$ and $y = g(\theta)$ where ' θ ' is a parameter then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ and $\frac{d^2y}{dx^2} =$

$$\frac{\frac{d}{d\theta} \left(\frac{dy}{dx} \right)}{\frac{dx}{d\theta}} \quad \text{In general } \frac{d^n y}{dx^n} = \frac{d}{d\theta} \left(\frac{d^{n-1} y}{dx^{n-1}} \right) \bigg/ \frac{dx}{d\theta}.$$

11. ANALYSIS AND GRAPHS OF SOME INVERSE TRIGONOMETRIC FUNCTIONS :

(a) $y = f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$

$$= \begin{cases} 2 \tan^{-1} x; & |x| \leq 1 \\ \pi - 2 \tan^{-1} x; & x > 1 \\ -(\pi + 2 \tan^{-1} x); & x < -1 \end{cases}$$

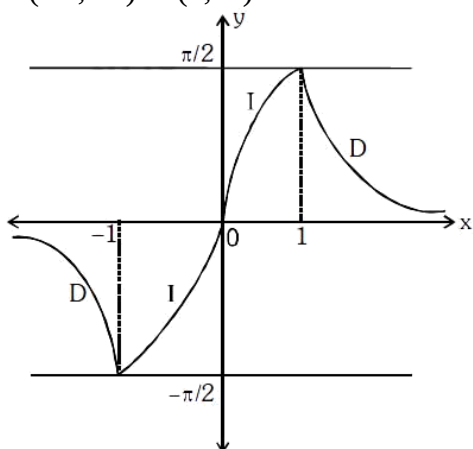
Important points :

- (i) Domain is $x \in \mathbb{R}$ and range is $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$

- (ii) f is continuous for all x but not differentiable at $x = 1, -1$

(iii) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } |x| < 1 \\ \text{non existent} & \text{for } |x| = 1 \\ \frac{-2}{1+x^2} & \text{for } |x| > 1 \end{cases}$

- (iv) Increasing in $(-1, 1)$ and Decreasing in $(-\infty, -1) \cup (1, \infty)$



(b) Consider $y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

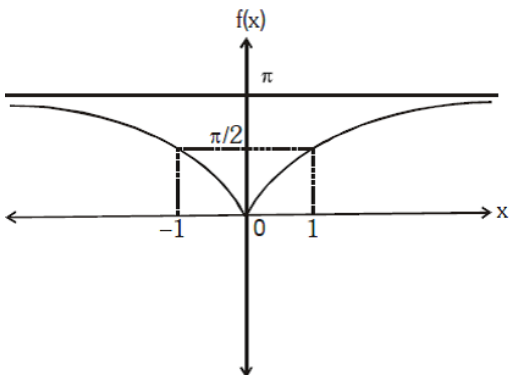
$$= \begin{cases} 2 \tan^{-1} x & \text{if } x \geq 0 \\ -2 \tan^{-1} x & \text{if } x < 0 \end{cases}$$

Important points :

- (i) Domain is $x \in \mathbb{R}$ & range is $[0, \pi)$
 (ii) Continuous for all x but not differentiable at $x = 0$

(iii) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } x > 0 \\ \text{non existent} & \text{for } x = 0 \\ -\frac{2}{1+x^2} & \text{for } x < 0 \end{cases}$

- (iv) Increasing in $(0, \infty)$ & Decreasing in $(-\infty, 0)$

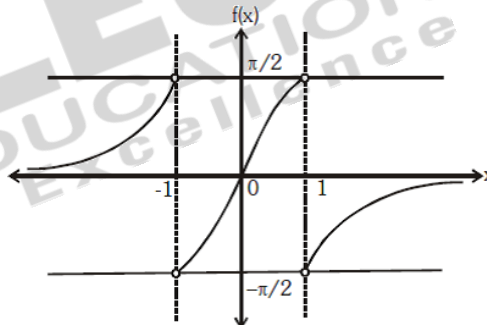


(c) $y = f(x) = \tan^{-1} \frac{2x}{1-x^2}$

$$= \begin{cases} 2 \tan^{-1} x & |x| < 1 \\ \pi + 2 \tan^{-1} x & x < -1 \\ -(\pi - 2 \tan^{-1} x) & x > 1 \end{cases}$$

Important points :

- (i) Domain is $\mathbb{R} - \{1, -1\}$ and range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (ii) It is neither continuous nor differentiable at $x = 1, -1$
 (iii) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & |x| \neq 1 \\ \text{non existent} & |x| = 1 \end{cases}$
 (iv) Increasing $\forall x$ in its domain
 (v) It is bounded for all x



(d) $y = f(x) = \sin^{-1}(3x - 4x^3)$

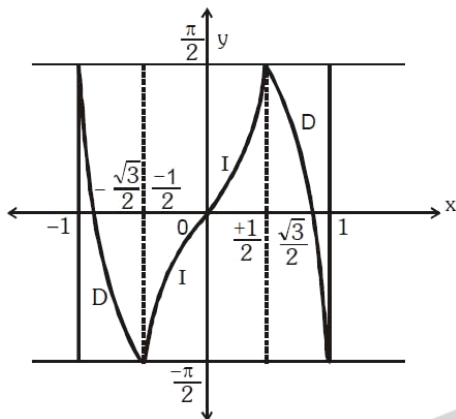
$$= \begin{cases} -(\pi + 3 \sin^{-1} x) & \text{if } -1 \leq x < -\frac{1}{2} \\ 3 \sin^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3 \sin^{-1} x & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

Important points :

- (i) Domain is $x \in [-1, 1]$ and range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
 (ii) Continuous everywhere in its domain
 (iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}$
 (iv) $\frac{dy}{dx}$

$$= \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$

- (v) Increasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and Decreasing in $\left[-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right]$



(e) $y = f(x) = \cos^{-1}(4x^3 - 3x)$

$$= \begin{cases} 3\cos^{-1}x - 2\pi & \text{if } -1 \leq x < -\frac{1}{2} \\ 2\pi - 3\cos^{-1}x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3\cos^{-1}x & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

Important points :

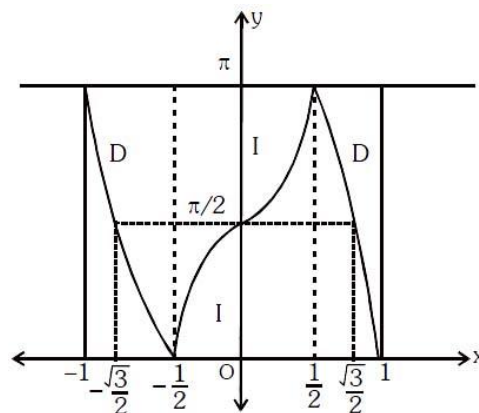
- (i) Domain is $x \in [-1, 1]$ and range is $[0, \pi]$
(ii) Continuous everywhere in its domain

- (iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}$

(iv) $\frac{dy}{dx}$

$$= \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$

- (v) Increasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and Decreasing in $\left[-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right]$



GENERAL NOTE :

Concavity is decided by the sign of 2nd derivative as :

$$\frac{d^2y}{dx^2} > 0 \Rightarrow \text{Concave upwards ; } \frac{d^2y}{dx^2} < 0 \Rightarrow \text{Concave downwards}$$

Illustration 17 :

Find the interval for which $f(x) = x^3 + x + 1$ is

- (i) concave upwards
(ii) concave downwards.

Solution :

$$f(x) = x^3 + x + 1$$

$$f'(x) = 3x^2 + 1$$

$$f''(x) = 6x$$

- (i) $f''(x) = 6x > 0 \Rightarrow \text{Concave upwards}$
 $\Rightarrow x \in (0, \infty)$
(ii) $f''(x) = 6x < 0 \Rightarrow \text{Concave downwards}$
 $\Rightarrow x \in (-\infty, 0)$ **Ans.**

Illustration 18 :

If $x = a(t + \sin t)$ and $y = a(1 - \cos t)$,

$$\text{find } \frac{d^2y}{dx^2}.$$

Solution :

Here $x = a(t + \sin t)$ and $y = a(1 - \cos t)$
Differentiating both sides w.r.t. t , we get :

$$\frac{dx}{dt} = a(1 + \cos t) \text{ and } \frac{dy}{dt} = a(\sin t)$$

$$\therefore \frac{dy}{dx} = \frac{a \sin t}{a(1 + \cos t)} = \frac{2 \sin \frac{t}{2} \cdot \cos \frac{t}{2}}{2 \cos^2 \frac{t}{2}} = \tan\left(\frac{t}{2}\right)$$

Again differentiating both sides, we get,

$$\frac{d^2y}{dx^2} = \sec^2\left(\frac{t}{2}\right) \cdot \frac{1}{2} \cdot \frac{dt}{dx} = \frac{1}{2} \sec^2(t/2).$$

$$\frac{1}{a(1 + \cos t)} = \frac{1}{2a} \cdot \frac{\sec^2\left(\frac{t}{2}\right)}{2 \cos^2 \frac{t}{2}}$$

$$\text{Hence, } \frac{d^2y}{dx^2} = \frac{1}{4a} \cdot \sec^4\left(\frac{t}{2}\right) \quad \text{Ans.}$$

Illustration 19 :

$y = f(x)$ and $x = g(y)$ are inverse functions of each other then express $g'(y)$ and $g''(y)$ in terms of derivative of $f(x)$.

Solution :

$$\frac{dy}{dx} = f'(x) \text{ and } \frac{dx}{dy} = g'(y)$$

$$\Rightarrow g'(y) = \frac{1}{f'(x)} \quad \text{.....(i)}$$

Again differentiating w.r.t. to y

$$g''(y) = \frac{d}{dy} \left(\frac{1}{f'(x)} \right) = \frac{d}{dx} \left(\frac{1}{f'(x)} \right) \cdot \frac{dx}{dy}$$

$$= -\frac{f''(x)}{(f'(x))^2} \cdot \left(\frac{1}{f'(x)} \right)$$

$$\Rightarrow g''(y) = -\frac{f''(x)}{(f'(x))^3} \quad \text{.....(ii)}$$

Which can also be remembered as

$$\frac{d^2x}{dy^2} = -\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3} \quad \text{Ans.}$$

Do yourself : 8

- (i) If $y = x e^{x^2}$ then find y'' .
- (ii) Find y'' at $x = \pi/4$, if $y = x \tan x$.
- (iii) Prove that the function $y = e^x \sin x$ satisfies the relationship $y'' - 2y' + 2y = 0$.

12. DIFFERENTIATION OF DETERMINANTS :

$$\text{If } F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}, \text{ where } f,$$

$g, h, \ell, m, n, u, v, w$ are differentiable functions of x then

$$F'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

Not : Sometimes it is better to expand the determinant first and then differentiate.

Illustration 20 :

$$\text{If } f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}, \text{ find } f'(x).$$

Solution :

$$\text{Here, } f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

on differentiating, we get,

$$\Rightarrow f'(x) = \begin{vmatrix} \frac{d}{dx}(x) & \frac{d}{dx}(x^2) & \frac{d}{dx}(x^3) \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

$$+ \begin{vmatrix} x & x^2 & x^3 \\ \frac{d}{dx}(1) & \frac{d}{dx}(2x) & \frac{d}{dx}(3x^2) \\ 0 & 2 & 6x \end{vmatrix}$$

$$+ \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ \frac{d}{dx}(0) & \frac{d}{dx}(2) & \frac{d}{dx}(6x) \end{vmatrix}$$

$$\text{or } f'(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 0 & 2 & 6x \\ 0 & 2 & 6x \end{vmatrix}$$

$$+ \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$$

As we know if any two rows or columns are equal, then value of determinant is zero.

$$= 0 + 0 + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$$

$$\therefore f'(x) = 6(2x^2 - x^2)$$

$$\text{Therefore, } f'(x) = 6x^2$$

Ans.

Do yourself : 9

(i) If $f(x) = \begin{vmatrix} e^x & x^2 \\ \ln x & \sin x \end{vmatrix}$, then find $f'(1)$.

(ii) If $f(x) = \begin{vmatrix} 2x & x^2 & x^3 \\ x^2 + 2x & 1 & 3x + 1 \\ 2x & 1 - 3x^2 & 5x \end{vmatrix}$, then find $f'(1)$.

13. L'HOPITAL'S RULE :

(a) This rule is applicable for the indeterminate forms of the type $\frac{0}{0}$, $\frac{\infty}{\infty}$.

If the function $f(x)$ and $g(x)$ are differentiable in certain neighborhood of the point 'a', except, may be, at the point 'a' itself and $g'(x) \neq 0$, and if

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \text{ or } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty, \text{ then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists (L' Hopital's rule). The point 'a' may be either finite or improper ($+\infty$ or $-\infty$).

(b) Indeterminate forms of the type $0 \cdot \infty$ or $\infty - \infty$ are reduced to forms of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ by algebraic transformations.

(c) Indeterminate forms of the type 1^∞ , ∞^0 or 0^0 are reduced to forms of the type $0 \times \infty$ by taking logarithms or by the transformation $[f(x)]^{\phi(x)} = e^{\phi(x) \cdot \ln f(x)}$.

Illustration 21 :

Evaluate $\lim_{x \rightarrow 0} |x|^{\sin x}$

Solution :

$$\lim_{x \rightarrow 0} |x|^{\sin x} = \lim_{x \rightarrow 0} e^{\sin x \log_e |x|} = e^{\lim_{x \rightarrow 0} \frac{\log_e |x|}{\csc x}}$$

$$= e^{\lim_{x \rightarrow 0} \frac{1/x}{-\csc x \cot x}}$$

(applying L'Hopital's rule)

$$e^{\lim_{x \rightarrow 0} \frac{-\sin^2 x}{x \cos x}} = e^{\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \left(\frac{x}{\cos x} \right)} = e^{-(1)^2 \cdot (0)}$$

$$= e^0 = 1$$

Ans.

Illustration 22 :

Solve $\lim_{x \rightarrow 0^+} \log_{\sin x} \sin 2x$.

Solution :

Here $\lim_{x \rightarrow 0^+} \log_{\sin x} \sin 2x$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0^+} \frac{\log \sin 2x}{\log \sin x} \quad \left(\frac{-\infty}{-\infty} \text{ form} \right) \\
 &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\sin 2x} \cdot 2 \cos 2x}{\frac{1}{\sin x} \cdot \cos x} \\
 &\quad \{ \text{applying L'Hopital's rule} \} \\
 &= \lim_{x \rightarrow 0^+} \frac{\left(\frac{2x}{\sin(2x)} \right) \cos 2x}{\left(\frac{x}{\sin x} \right) \cos x} = \lim_{x \rightarrow 0^+} \frac{\cos 2x}{\cos x} \\
 &= 1 \quad \text{Ans.}
 \end{aligned}$$

Illustration 23 :

Evaluate $\lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n}$.

Solution :

Here, $A = \lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n}$ (∞^0 form)

$$\begin{aligned}
 \therefore \log A &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{e^n}{\pi} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{n \log e - \log \pi}{n} \quad \left(\frac{\infty}{\infty} \text{ form} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{\log e - 0}{1} \\
 &\quad \{ \text{applying L'Hopital's rule} \} \\
 \log A &= 1
 \end{aligned}$$

$$\Rightarrow A = e^1 \text{ or } \lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n} = e$$

Do yourself : 10

(i) Using L'Hopital's rule find

(a) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ (b) $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$

(ii) Using L'Hopital's rule verify that :

(a) $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} = -\frac{1}{2}$

(b) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

Miscellaneous Illustration :

Illustration 24 :

Find second order derivative of $y = \sin x$ with respect to $z = e^x$.

Solution :

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{dy/dx}{dz/dx} = \frac{\cos x}{e^x} \\
 \Rightarrow \frac{d^2y}{dz^2} &= \frac{d}{dx} \left(\frac{\cos x}{e^x} \right) \cdot \frac{dx}{dz} \\
 &= \frac{-e^x \sin x - \cos x e^x}{(e^x)^2} \cdot \frac{1}{e^x} \\
 \Rightarrow \frac{d^2y}{dz^2} &= -\frac{(\sin x + \cos x)}{e^{2x}}
 \end{aligned}$$

Illustration 25 :

Let a function f satisfies

$$\begin{aligned}
 f\left(\frac{x+y}{2}\right) &= \frac{f(x) + f(y)}{2} \quad \forall x, y \in \mathbb{R} \text{ and} \\
 f'(0) &= a, f(0) = b, \text{ then find } f(x) \text{ hence} \\
 &\text{find } f''(x).
 \end{aligned}$$

Solution :

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2}$$

Diff. w.r.t. 'x'

$$f'\left(\frac{x+y}{2}\right) \cdot \frac{1}{2} = \frac{1}{2} f'(x)$$

$$\left(\because x \text{ \& } y \text{ are independent to each other, } \therefore \frac{dy}{dx} = 0 \right)$$

$$f'\left(\frac{x+y}{2}\right) = f'(x)$$

Let $x = 0$ and $y = x$

$$f'\left(\frac{x}{2}\right) = f'(0) = a$$

$$\Rightarrow f'(x) = a$$

On integrating, we get

$$f(x) = ax + b \quad (\because f(0) = b)$$

$$\Rightarrow f''(x) = 0$$

Illustration 26:

Prove that $\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \frac{1}{2^6} \sec^2 \frac{x}{2^3} + \dots \infty = \operatorname{cosec}^2 x - \frac{1}{x^2}$

Solution :

Let $\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \infty$

$= \lim_{n \rightarrow \infty} \cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \cos \frac{x}{2^n}$

$\Rightarrow \lim_{n \rightarrow \infty} \frac{\sin x}{2^n \sin \frac{x}{2^n}} = \frac{\sin x}{x} \Rightarrow \cos \frac{x}{2} \cdot \cot \frac{x}{2^2}$

$\cos \frac{x}{2^3} \dots \infty = \frac{\sin x}{x}$

$\Rightarrow \ln \left(\cos \frac{x}{2} \right) + \ln \left(\cos \frac{x}{2^2} \right) + \ln$

$\left(\cos \frac{x}{2^3} \right) + \dots \infty = \ln \sin x - \ln x$

Diff. w.r.t.x

$-\left(\frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \frac{x}{2^2} + \dots \infty \right)$

$= \cot x - \frac{1}{x}$

Diff. w.r.t.x again

$\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \frac{1}{2^6} \sec^2 \frac{x}{2^3} +$

$\dots \infty = \operatorname{cosec}^2 x - \frac{1}{x^2}$

Hence proved

ANSWERS FOR DO YOURSELF

1 : (i)(a) $\frac{1}{x}$ (b) $-\frac{1}{x^2}$

2 : (i) (a) $3x^2 + 12x + 11$
(b) $5e^{5x} \tan(x^2 + 2) + 2xe^{5x} \sec^2(x^2 + 2)$

3 : (i) $x^x (\ln x + 1)$
(ii) $y(1 + 2x + 3x^2 + 4x^3)$

4. (i) -1 (ii) $\frac{1}{t}$
(iii) $2(x^{\ln x})(\ln x)$

5 : (i) $\frac{\cos(x-y)-1}{\cos(x-y)+1}$
(ii) $y' = -\left(\frac{2x+e^y}{xe^y+1} \right), -1$

6 : (i) (a) -6 (b) 3
(c) -6

7 : (i) D

8 : (i) $y'' = 4y + 2xy'$
(ii) $\pi + 4$

9 : (i) $e(\sin 1 + \cos 1) - 1$ (ii) 9

10. (i) (a) $\frac{1}{3}$ (b) $\frac{1}{2}$

Result JEE Main

Result Highlights 2019



100 Percentile
HIMANSHU GAURAV SINGH
2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



99.98 Percentile
SARTHAK ROUT
2020 (CCP)



99.98 Percentile
VIBHAV AGGARWAL
2019 (CCP)



99.97 Percentile
RITVIK GUPTA
2020 (DLP)



99.97 Percentile
BHAVYA JAIN
2020 (CCP)



99.96 Percentile
AYUSH PATTNAIK
2019 (CCP)



99.96 Percentile
SAYANTAN DHAR
2020 (DLP)

Result Highlights 2018

AIR
29



Sarthak Behera

AIR
36



Shashank Agrawal

AIR
46



Shreya Pathak

AIR
48



Raaghav Raaj

AIR
2
SC



Aayush Kadam

AIR
18
OBC



Gunjan

Result Highlights 2017

AIR
12



Abhay Goyal

AIR
14



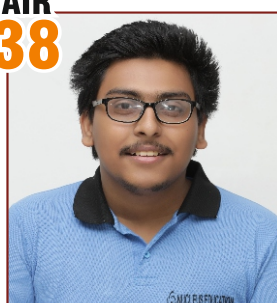
Piyush Tibarwal

AIR
20



Abhinav Kumar

AIR
38



Tushar Gautam

AIR
49



Saurav Sahoo

AIR
60



Pratyush

Mentored by Team

AIR
4



Kushal Babel
2014

AIR
15



Gourika Bang
2015

AIR
21



Amit Malav
2013

AIR
24



Suraj Sanjay Jog
2012

AIR
38



Chitraang Murdia
2014

AIR
58



Pulkit Tandon
2012

AIR
68



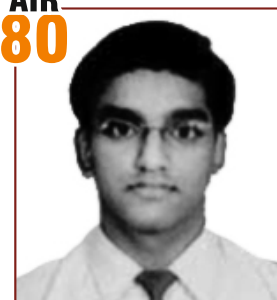
Kapil Shobhnani
2016

AIR
70



Utkarsh Sinha
2015

AIR
80



S John S
2015

IIT MATHEMATICS

METHODS OF DIFFERENTIATION



RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
Gen. Category - 2014



AIR-3
GOVIND LAHOTI
Gen. Category - 2014

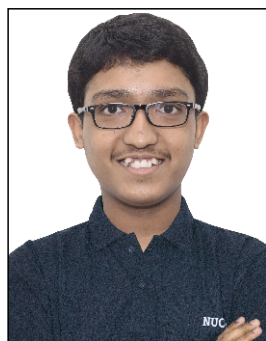


AIR-6
NISHIT AGARWAL
Gen. Category - 2012

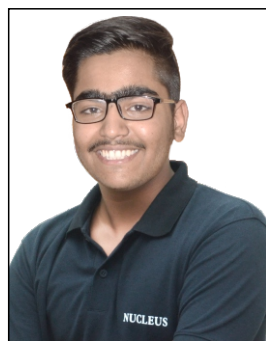


AIR-8
AMEY GUPTA
Gen. Category - 2014

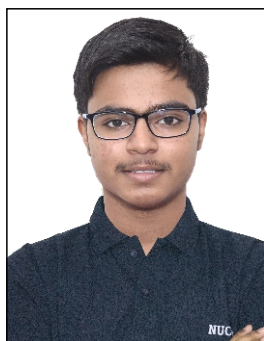
JEE MAIN RESULTS 2020 (January Attempt) OF NUCLEUS EDUCATION



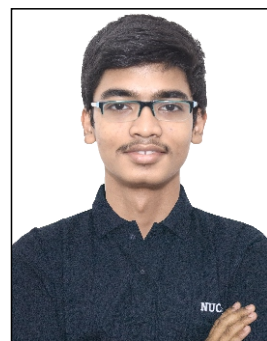
100 Percentile
(Physics)
DAKSH Khandelwal
2020



100 Percentile
(Maths & Physics)
VAIBHAV SAHA
2020



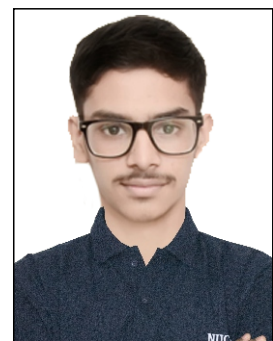
100 Percentile
(Physics)
ANISH MOHAN
2020



100 Percentile
(Maths)
ARCHIT PATNAIK
2020

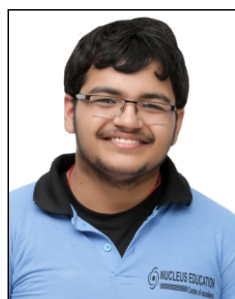


100 Percentile
(Maths)
SWAPNIL YASASVI
2020

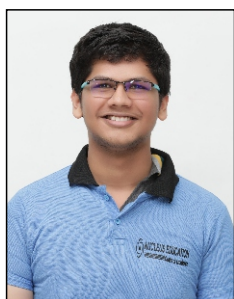


100 Percentile
(Maths)
PARSHANT ARORA
2020

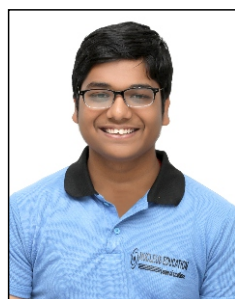
JEE ADVANCED RESULTS OF NUCLEUS EDUCATION



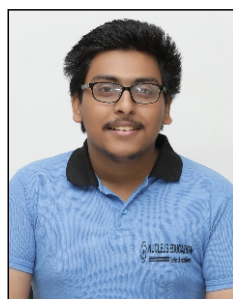
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LAKSHAY SHARMA
Gen. Category - 2017



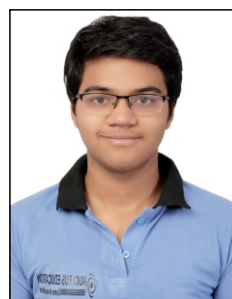
AIR-12
YATEESH AGRAWAL
Gen. Category - 2017



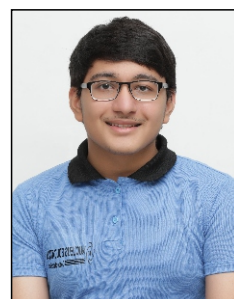
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ABHEY GOYAL
Gen. Category - 2017



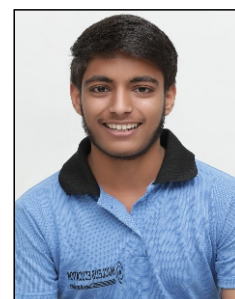
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TUSHAR GAUTAM
Gen. Category - 2017



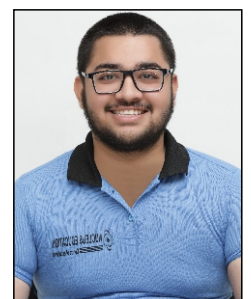
AIR-37
PIYUSH TIBAREWAL
Gen. Category - 2017



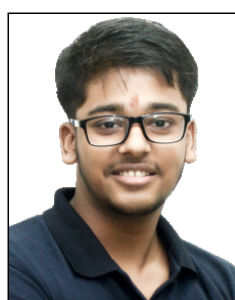
AIR-42
SATVIK
Gen. Category - 2017



AIR-66
MAYANK DUBEY
Gen. Category - 2017



AIR-98
HRITHIK
Gen. Category - 2017



AIR-20
SHASHANK AGRAWAL
Gen. Category - 2018



AIR-27
RAAGHAV RAAJ
Gen. Category - 2018



AIR-32
SHREYA PATHAK
Gen. Category - 2018



AIR-61
SIDDHANT CHOUDHARY
Gen. Category - 2018



AIR-67
ANISWAR S KRISHNAN
Gen. Category - 2018
DLP



AIR-78
AAYUSH KADAM
Gen. Category - 2018



AIR-61
SARTHAK BEHERA
Gen. Category - 2018
*SDCCP



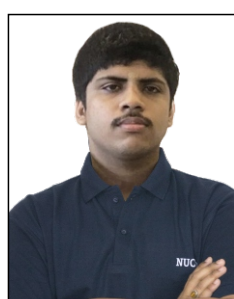
AIR-91
ANDREWS G. VARGHESE
Gen. Category - 2018
*SDCCP



AIR-2
HIMANSHU GAURAV SINGH
Gen. Category - 2019
*SDCCP



AIR-19
VIBHAV AGGARWAL
Gen. Category - 2019



AIR-33
S. PRAJEETH
Gen. Category - 2019
*SDCCP



AIR-48
SOHAM MISTRI
Gen. Category - 2019
DLP



AIR-51
SAYANTAN DHAR
Gen. Category - 2019
DLP



AIR-53
GAURAV KRISHAN GUPTA
Gen. Category - 2019
DLP



AIR-86
SATVIK JAIN
Gen. Category - 2019

EXERCISE # 1

1. If $y = \frac{1}{1+x^{\beta-\alpha}+x^{\gamma-\alpha}} + \frac{1}{1+x^{\alpha-\beta}+x^{\gamma-\beta}} + \frac{1}{1+x^{\alpha-\gamma}+x^{\beta-\gamma}}$, then $\frac{dy}{dx}$ is equal to-
 (A) 0 (B) 1
 (C) $(\alpha + \beta + \gamma)x^{\alpha+\beta+\gamma-1}$ (D) $\alpha\beta\gamma$
2. If $f(x) = |\cos x|$, then $f'\left(\frac{3\pi}{4}\right)$ is equal to -
 (A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) -1
3. $\frac{d}{dx}(e^x \sin \sqrt{3} x)$ equals-
 (A) $e^x \sin(\sqrt{3} x + \pi/3)$ (B) $2e^x \sin(\sqrt{3} x + \pi/3)$
 (C) $\frac{1}{2} e^x \sin(\sqrt{3} x + \pi/3)$ (D) $\frac{1}{2} e^x \sin(\sqrt{3} x - \pi/3)$
4. $\frac{d}{dx}(\ln \sin \sqrt{x})$ is equal to-
 (A) $\frac{\tan \sqrt{x}}{2\sqrt{x}}$ (B) $\frac{\cot \sqrt{x}}{2\sqrt{x}}$ (C) $\frac{\cot \sqrt{x}}{2x}$ (D) $\frac{\cot \sqrt{x}}{2\sqrt{x}}$
5. If $y = \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{dx}$ equals -
 (A) $\frac{y}{1-x^2}$ (B) $\frac{y}{x^2-1}$ (C) $\frac{y}{1+x^2}$ (D) $\frac{y}{y^2-1}$
6. If $y = \ln \left\{ \frac{x + \sqrt{a^2 + x^2}}{a} \right\}$, then the value of $\frac{dy}{dx}$ is-
 (A) $\sqrt{a^2 - x^2}$ (B) $a\sqrt{a^2 + x^2}$ (C) $\frac{1}{\sqrt{a^2 + x^2}}$ (D) $x\sqrt{a^2 + x^2}$
7. If $x = y \ln(xy)$, then $\frac{dx}{dy}$ equals-
 (A) $\frac{y(x-y)}{x(x+y)}$ (B) $\frac{x(x+y)}{y(x-y)}$ (C) $\frac{y(x+y)}{x(x-y)}$ (D) $\frac{x(x-y)}{y(x+y)}$

8. If $(\cos x)^y = (\sin y)^x$, then $\frac{dy}{dx}$ equals-
- (A) $\frac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$ (B) $\frac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$
 (C) $\frac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$ (D) $\frac{\log \sin y + y \tan x}{\log \cos y - y \cot x}$
9. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to-
- (A) $\frac{2^x + 2^y}{2^x - 2^y}$ (B) $\frac{2^x + 2^y}{1 + 2^{x+y}}$ (C) $2^{x-y} \left(\frac{2^y - 1}{1 - 2^x} \right)$ (D) $\frac{2^x + y - 2^x}{2^y}$
10. If $x = a(t - \sin t)$, $y = a(1 + \cos t)$, then $\frac{dy}{dx}$ equals-
- (A) $-\tan \frac{t}{2}$ (B) $\cot \frac{t}{2}$ (C) $-\cot \frac{t}{2}$ (D) $\tan \frac{t}{2}$
11. The differential coefficient of $\sec^{-1} \left(\frac{1}{2x^2 - 1} \right)$ w.r.t. $\sqrt{1 - x^2}$ is-
- (A) $1/x^2$ (B) $2/x^3$ (C) $x/2$ (D) $2/x$
12. If $x^3 - y^3 + 3xy^2 - 3x^2y + 1 = 0$, then at $(0, 1)$ $\frac{dy}{dx}$ equals-
- (A) 1 (B) -1 (C) 2 (D) 0
13. $\frac{d}{d\theta} \left(\tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) \right)$ equals, if $-\pi < \theta < \pi$ -
- (A) $1/2$ (B) 1 (C) $\sec \theta$ (D) $\operatorname{cosec} \theta$
14. $\frac{d}{dx} \cot^{-1} \left(\frac{1+x}{1-x} \right)$ is equal to, if $x > -1$
- (A) $\frac{1}{1+x^2}$ (B) $\frac{1}{1-x^2}$ (C) $-\frac{1}{1+x^2}$ (D) $-\frac{1}{1-x^2}$
15. If $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, then $\frac{dy}{dx}$ is equal to-
- (A) 1 (B) 0 (C) -1 (D) -2

16. $\frac{d}{dx} \left(\tan^{-1} \left(\frac{\sqrt{x} - x}{1 + x^{3/2}} \right) \right)$ equals- ($x \geq 0$)

(A) $\frac{1}{2\sqrt{x}(1+x)} - \frac{1}{1+x^2}$

(B) $\frac{1}{2\sqrt{x}(1+x)} + \frac{1}{1+x^2}$

(C) $\frac{1}{1+x} - \frac{1}{1+x^2}$

(D) $\frac{1}{1+x} + \frac{1}{1+x^2}$

17. If g is the inverse of f and $f'(x) = \frac{1}{1+x^3}$ then $g'(x)$ is equal to-

(A) $1 + [g(x)]^3$

(B) $\frac{-1}{2(1+x^2)}$

(C) $\frac{1}{2(1+x^2)}$

(D) $\frac{1}{1+[g(x)]^3}$

18. If $x^2 + y^2 = 1$, then-

(A) $yy'' - 2(y')^2 + 1 = 0$

(B) $yy'' + (y')^2 + 1 = 0$

(C) $yy'' + (y')^2 - 1 = 0$

(D) $yy'' + 2(y')^2 + 1 = 0$

19. Let f be a function defined for all $x \in \mathbb{R}$. If f is differentiable and $f(x^3) = x^5$ for all $x \in \mathbb{R}$ ($x \neq 0$), then the value of $f'(27)$ is-

(A) 15

(B) 45

(C) 0

(D) 35

20. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are given, then -

(A) $\Delta_1 = 3(\Delta_2)^2$

(B) $\frac{d}{dx} \Delta_1 = 3\Delta_2$

(C) $\frac{d}{dx} \Delta_1 = 3(\Delta_2)^2$

(D) $\Delta_1 = 3(\Delta_2)^{3/2}$

21. Let $y = x^3 - 8x + 7$ and $x = f(t)$. If $\frac{dy}{dt} = 2$ and $x = 3$ at $t = 0$, then $\frac{dx}{dt}$ at $t = 0$ is given by :

(A) 1

(B) $\frac{19}{2}$

(C) $\frac{2}{19}$

(D) None of these

22. If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$ is equal to :

(A) 1

(B) -1

(C) $\frac{1}{\sqrt{2}}$

(D) $-\frac{1}{\sqrt{2}}$

23. If $y = \sin^{-1} \frac{x^2 - 1}{x^2 + 1} + \sec^{-1} \frac{x^2 + 1}{x^2 - 1}$, $|x| > 1$, then $\frac{dy}{dx}$ is equal to :
- (A) $\frac{x}{x^4 - 1}$ (B) $\frac{x^2}{x^4 - 1}$ (C) 0 (D) 1
24. If $y = x - x^2$, then the derivative of y^2 w.r.t. x^2 is :
- (A) $2x^2 + 3x - 1$ (B) $2x^2 - 3x + 1$ (C) $2x^2 + 3x + 1$ (D) none of these
25. Let $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$. Then $\lim_{x \rightarrow 0} \frac{f'(x)}{x} =$
- (A) 2 (B) -2 (C) -1 (D) 1
26. Let $e^{f(x)} = \ln x$. If $g(x)$ is the inverse function of $f(x)$ then $g'(x)$ equal to :
- (A) e^x (B) $e^x + x$ (C) e^{x+e^x} (D) $e^{x + \ln x}$
27. Consider $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in A.P., then $f'(a), f'(b)$ and $f'(c)$ are in :
- (A) G.P. (B) H.P. (C) A.G.P. (D) A.P.
28. If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2 f(x)$, then $\frac{dy}{dx}$ at $x = -1$ is equal to :
- (A) 0 (B) $\frac{1}{14}$ (C) $-\frac{1}{14}$ (D) none of these
29. Let $f(x)$ be a polynomial in x . Then the second derivative of $f(e^x)$ w.r.t. x is :
- (A) $f''(e^x) \cdot e^x + f'(e^x)$ (B) $f''(e^x) \cdot e^{2x} + f'(e^x) \cdot e^{2x}$
(C) $f''(e^x) \cdot e^{2x}$ (D) $f''(e^x) \cdot e^{2x} + f'(e^x) \cdot e^x$
30. Let $y = x^{x+1} + \left(1 + \frac{1}{x}\right)^x$ then $y'(1)$ equals
- (A) $(\ln 2) + 1$ (B) $(2 \ln 2) + 1$ (C) $(\ln 2) - 1$ (D) $(2 \ln 2) - 1$
31. Let $u(x)$ and $v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)}\right)' = q$, then $\frac{p+q}{p-q}$ has the value equal to -
- (A) 1 (B) 0 (C) 7 (D) -7

32. Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to
(A) 19 (B) 9 (C) 17 (D) 14
33. If $f(x) = \begin{cases} x \tan^{-1} x + \sec^{-1}(1/x), & x \in (-1, 1) - \{0\} \\ \pi/2, & \text{if } x = 0 \end{cases}$, then $f'(0)$ is -
(A) equal to -1 (B) equal to 0 (C) equal to 1 (D) non existent
34. Given: $f(x) = 4x^3 - 6x^2 \cos 2a + 3x \sin 2a \cdot \sin 6a + \sqrt{\ln(2a - a^2)}$ then
(A) $f(x)$ is not defined at $x = 1/2$ (B) $f'(1/2) < 0$
(C) $f'(x)$ is not defined at $x = 1/2$ (D) $f'(1/2) > 0$
35. If $x = t^3 + t + 5$ & $y = \sin t$, then $\frac{d^2y}{dx^2} =$
(A) $-\frac{(3t^2+1)\sin t + 6t \cos t}{(3t^2+1)^3}$ (B) $\frac{(3t^2+1)\sin t + 6t \cos t}{(3t^2+1)^2}$
(C) $-\frac{(3t^2+1)\sin t + 6t \cos t}{(3t^2+1)^2}$ (D) $\frac{\cos t}{3t^2+1}$
36. If $f(x) = \frac{a + \sqrt{a^2 - x^2} + x}{\sqrt{a^2 - x^2} + a - x}$ where $a > 0$ and $x < a$, then $f'(0)$ has the value equal to -
(A) $\sqrt{a}$ (B) a (C) $\frac{1}{\sqrt{a}}$ (D) $\frac{1}{a}$
37. Suppose that $f(0) = 0$ and $f'(0) = 2$, and let $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ is equal to
(A) 0 (B) 1 (C) 6 (D) 8
38. If f is differentiable in $(0, 6)$ & $f'(4) = 5$, then $\lim_{x \rightarrow 2} \frac{f(4) - f(x^2)}{2 - x} =$
(A) 5 (B) 5/4 (C) 10 (D) 20
39. If $y = x + e^x$ then $\frac{d^2x}{dy^2}$ is :
(A) e^x (B) $-\frac{e^x}{(1+e^x)^3}$ (C) $-\frac{e^x}{(1+e^x)^2}$ (D) $\frac{-1}{(1+e^x)^3}$

40. If f is twice differentiable such that

$$\begin{aligned} f''(x) &= -f(x), f'(x) = g(x) \\ h'(x) &= [f(x)]^2 + [g(x)]^2 \text{ and} \\ h(0) &= 2, h(1) = 4 \end{aligned}$$

then the equation $y = h(x)$ represents :

- (A) a curve of degree 2
- (B) a curve passing through the origin
- (C) a straight line with slope 2
- (D) a straight line with y intercept equal to -2 .

41. If $y = (A + Bx)e^{mx} + (m-1)^{-2}e^x$ then $\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ is equal to -

- (A) e^x
- (B) e^{mx}
- (C) e^{-mx}
- (D) $e^{(1-m)x}$

42. If $\frac{d^2x}{dy^2}\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} = K$ then the value of K is equal to

- (A) 1
- (B) -1
- (C) 2
- (D) 0

43. Let $f(x)$ be differentiable at $x = h$, then $\lim_{x \rightarrow h} \frac{(x+h)f(x) - 2hf(h)}{x-h}$ is equal to -

- (A) $f(h) + 2hf'(h)$
- (B) $2f(h) + hf'(h)$
- (C) $hf(h) + 2f'(h)$
- (D) $hf(h) - 2f'(h)$

44. If $y = at^2 + 2bt + c$ and $t = ax^2 + 2bx + c$, then $\frac{d^3y}{dx^3}$ equals

- (A) $24a^2(at+b)$
- (B) $24a(ax+b)^2$
- (C) $24a(at+b)^2$
- (D) $24a^2(ax+b)$

45. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f . The value of $g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ has value equal to

- (A) $\sqrt{2} - 1$
- (B) $\frac{\sqrt{2}+1}{\sqrt{2}}$
- (C) $2 - \sqrt{2}$
- (D) $\sqrt{2} + 1$

EXERCISE # 2

1. Let f , g and h are differentiable functions. If $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are $(f g)'(0) = 6$; $(g h)'(0) = 4$ and $(h f)'(0) = 5$ then compute the value of $(fgh)'(0)$.
2. (a) If $y = (\cos x)^{\ln x} + (\ln x)^x$ find $\frac{dy}{dx}$ (b) If $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$ Find $\frac{dy}{dx}$.
3. If $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln \sqrt{x+\sqrt{x^2+1}}$ prove that $2y = xy' + \ln y'$. Where y' denotes the derivative of y .
4. If $y = \ln \left(x^{e^x} \cdot a^{y^x} \right)^{y^x}$ find $\frac{dy}{dx}$.
5. If $y = 1 + \frac{x_1}{x-x_1} + \frac{x_2 \cdot x}{(x-x_1)(x-x_2)} + \frac{x_3 \cdot x^2}{(x-x_1)(x-x_2)(x-x_3)} + \dots$ upto $(n+1)$ terms then prove that $\frac{dy}{dx} = \frac{y}{x} \left[\frac{x_1}{x_1-x} + \frac{x_2}{x_2-x} + \frac{x_3}{x_3-x} + \dots + \frac{x_n}{x_n-x} \right]$
6. If $x = \operatorname{cosec} \theta - \sin \theta$; $y = \operatorname{cosec}^n \theta - \sin^n \theta$, then show that $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - n^2(y^2 + 4) = 0$.
7. If a curve is represented parametrically by the equations

$$x = \sin \left(t + \frac{7\pi}{12} \right) + \sin \left(t - \frac{\pi}{12} \right) + \sin \left(t + \frac{3\pi}{12} \right),$$

$$y = \cos \left(t + \frac{7\pi}{12} \right) + \cos \left(t - \frac{\pi}{12} \right) + \cos \left(t + \frac{3\pi}{12} \right)$$
 then find the value of $\frac{d}{dt} \left(\frac{x}{y} - \frac{y}{x} \right)$ at $t = \frac{\pi}{8}$.
8. If $x = \frac{1 + \ln t}{t^2}$ and $y = \frac{3 + 2 \ln t}{t}$. Show that $y \frac{dy}{dx} = 2x \left(\frac{dy}{dx} \right)^2 + 1$.
9. Differentiate $\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ w.r.t. $\sqrt{1-x^4}$.

10. Let $g(x)$ be a polynomial, of degree one & $f(x)$ be defined by $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$

Find the continuous function $f(x)$ satisfying $f'(1) = f(-1)$.

11. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 \cdot (x^3 - y^3)$, prove that $\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$.

12. If $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$, prove that $\frac{dy}{dx} = \frac{1}{2 - \frac{x}{x + \frac{1}{x + \frac{1}{x + \dots}}}}$

13. Let $f(x) = x + \frac{1}{2x} + \frac{1}{2x} + \frac{1}{2x} + \dots + \infty$. Compute the value of $f(100)$. $f'(100)$.

14. Find the derivative with respect to x of the function :

$$(\log_{\cos x} \sin x) (\log_{\sin x} \cos x)^{-1} + \arcsin \frac{2x}{1+x^2} \text{ at } x = \frac{\pi}{4}.$$

15. Suppose $f(x) = \tan(\sin^{-1}(2x))$

- (a) Find the domain and range of f .
(b) Express $f(x)$ as an algebraic function of x .
(c) Find $f'(1/4)$

16. (a) Let $f(x) = x^2 - 4x - 3$, $x > 2$ and let g be the inverse of f . Find the value of g' where $f(x) = 2$.
(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and $g(x) = f^{-1}(x)$, then find $g'(-1)$.

- (c) Suppose f^{-1} is the inverse function of a differentiable function f and let $G(x) = \frac{1}{f^{-1}(x)}$.

If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.

17. If $y = \tan^{-1} \frac{1}{x^2 + x + 1} + \tan^{-1} \frac{1}{x^2 + 3x + 3} + \tan^{-1} \frac{1}{x^2 + 5x + 7} + \tan^{-1} \frac{1}{x^2 + 7x + 13} + \dots$ to n terms. Find dy/dx , expressing your answer in 2 terms.
18. If $y = \tan^{-1} \frac{u}{\sqrt{1-u^2}}$ and $x = \sec^{-1} \frac{1}{2u^2 - 1}$, $u \in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ prove that $2 \frac{dy}{dx} + 1 = 0$.
19. If $y = \cot^{-1} \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$, find $\frac{dy}{dx}$ if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$.
20. If $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$, then $f'(x) = \lambda \cdot \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$. Find the value of λ .
21. (a) If $y = y(x)$ and it follows the relation $e^{xy} + y \cos x = 2$, then find (i) $y'(0)$ and (ii) $y''(0)$.
 (b) A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions
 $f(0) = 2$; $f'(0) = -5$ and $f''(0) = 3$.
 The function $g(x)$ is defined by $g(x) = e^{ax} + f(x) \forall x \in \mathbb{R}$, where 'a' is any constant.
 If $g'(0) + g''(0) = 0$. Find the value(s) of 'a'.
22. If $x = 2\cos t - \cos 2t$ & $y = 2\sin t - \sin 2t$, find the value of (d^2y/dx^2) when $t = (\pi/2)$.
23. Find the value of the expression $y^3 \frac{d^2y}{dx^2}$ on the ellipse $3x^2 + 4y^2 = 12$.
24. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ then find $f'(x)$.
25. Let $P(x)$ be a polynomial of degree 4 such that $P(1) = P(3) = P(5) = P'(7) = 0$. If the real number $x \neq 1, 3, 5$ is such that $P(x) = 0$ can be expressed as $x = p/q$ where 'p' and 'q' are relatively prime, then $(p + q)$ equals.

EXERCISE # 3

Evaluate the following limits using L'Hopital's Rule or otherwise (Q. No. 1 to 5) :

1. $\lim_{x \rightarrow 0} \left[\frac{1}{x \sin^{-1} x} - \frac{1-x^2}{x^2} \right]$
2. $\lim_{x \rightarrow 0} \frac{x + \ln(\sqrt{x^2+1} - x)}{x^3}$
3. $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right]$
4. $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1-x)}{x \cdot \tan^2 x}$
5. $\lim_{x \rightarrow 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 \cdot (\sin x)^{6000}}$
6. If $\lim_{x \rightarrow 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x \dots \cos nx}{x^2}$ has the value equal to 253, find the value of n (where $n \in \mathbb{N}$)
7. Given a real valued function f(x) as follows :
 $f(x) = \frac{x^2 + 2 \cos x - 2}{x^4}$ for $x < 0$; $f(0) = \frac{1}{12}$ and $f(x) = \frac{\sin x - \ln(e^x \cos x)}{6x^2}$ for $x > 0$. Test the continuity and differentiability of f(x) at $x = 0$.
8. Let $a_1 > a_2 > a_3 \dots a_n > 1$; $p_1 > p_2 > p_3 \dots p_n > 0$; such that $p_1 + p_2 + p_3 + \dots + p_n = 1$. Also $F(x) = (p_1 a_1^x + p_2 a_2^x + \dots + p_n a_n^x)^{1/x}$. Compute
 (a) $\lim_{x \rightarrow 0^+} F(x)$ (b) $\lim_{x \rightarrow \infty} F(x)$ (c) $\lim_{x \rightarrow -\infty} F(x)$
9. If $x_1, x_2, x_3, \dots, x_{n-1}$ be n zero's of the polynomial $P(x) = x^n + \alpha x + \beta$, where $x_i \neq x_j \forall i \neq j = 1, 2, 3, \dots, (n-1)$. Prove that the value of $Q(x) = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) \dots (x_1 - x_{n-1})$, is equal to ${}^nC_2 x_1^{n-2}$.
10. **Column-I** contains function defined on R and **Column-II** contains their properties. Match them:-

Column - I

- (A) $\lim_{n \rightarrow \infty} \left(\frac{1 + \tan \frac{\pi}{2n}}{1 + \sin \frac{\pi}{3n}} \right)^n$ equal
- (B) $\lim_{x \rightarrow 0^+} \frac{1}{(1 + \operatorname{cosec} x)^{\frac{1}{\ln(\sin x)}}}$ equals
- (C) $\lim_{x \rightarrow 0} \left(\frac{2}{\pi} \cos^{-1} x \right)^{1/x}$ equals

Column - II

- (P) e
- (Q) e^2
- (R) $e^{-2/\pi}$
- (S) $e^{\pi/6}$

EXERCISE # 4 (JM)

1. If $x^m \cdot y^n = (x + y)^{m+n}$, then $\frac{dy}{dx}$ is – [AIEEE-2006]
 (1) $\frac{x+y}{xy}$ (2) xy (3) $\frac{x}{y}$ (4) $\frac{y}{x}$
2. Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. then $y'(1)$ equals :- [AIEEE-2009]
 (1) $\log 2$ (2) $-\log 2$ (3) -1 (4) 1
3. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$.
 Let $g(x) = [f(2f(x) + 2)]^2$. Then $g'(0)$:- [AIEEE-2010]
 (1) 4 (2) -4 (3) 0 (4) -2
4. $\frac{d^2x}{dy^2}$ equals :- [AIEEE-2011]
 (1) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (2) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$ (3) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (4) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$
5. If $y = \sec(\tan^{-1}x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to : [JEE-MAIN-2013]
 (1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$ (3) 1 (4) $\sqrt{2}$
6. If g is the inverse of a function f and $f'(x) = \frac{1}{1+x^5}$, then $g'(x)$ is equal to : [JEE-MAIN-2014]
 (1) $1+x^5$ (2) $5x^4$ (3) $\frac{1}{1+\{g(x)\}^5}$ (4) $1+\{g(x)\}^5$
7. If for $x \in \mathbb{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then : [JEE(Main)-2016]
 (1) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$
 (2) g is not differentiable at $x = 0$
 (3) $g'(0) = \cos(\log 2)$
 (4) $g'(0) = -\cos(\log 2)$
8. If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x} \cdot g(x)$ then $g(x)$ equals : [JEE (Main)2017]
 (1) $\frac{3}{1+9x^3}$ (2) $\frac{9}{1+9x^3}$ (3) $\frac{3x\sqrt{x}}{1-9x^3}$ (4) $\frac{3x}{1-9x^3}$

9. If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is : **[JEE (Main) 2019]**
- (1) $\frac{3}{2\sqrt{2}}$ (2) $\frac{1}{6\sqrt{2}}$ (3) $\frac{1}{3\sqrt{2}}$ (4) $\frac{1}{6}$
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then $f(2)$ equals: **[JEE (Main) 2019]**
- (1) 8 (2) 30 (3) -4 (4) -2
11. If $x \log_e (\log_e x) - x^2 + y^2 = 4$ ($y > 0$), then $\frac{dy}{dx}$ at $x = e$ is equal to : **[JEE (Main) 2019]**
- (1) $\frac{e}{\sqrt{4+e^2}}$ (2) $\frac{(2e-1)}{2\sqrt{4+e^2}}$ (3) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (4) $\frac{(1+2e)}{\sqrt{4+e^2}}$
12. For $x > 1$, if $(2x)^{2y} = 4e^{2x-2y}$, then $(1 + \log_e 2x)^2 \frac{dx}{dy}$ is equal to : **[JEE (Main) 2019]**
- (1) $\frac{x \log_e 2x - \log_e 2}{x}$ (2) $x \log_e 2x$
(3) $\log_e 2x$ (4) $\frac{x \log_e 2x + \log_e 2}{x}$
13. If $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2$, $x \in \left(0, \frac{\pi}{2} \right)$, then $\frac{dy}{dx}$ is equal to **[JEE (Main) 2019]**
- (1) $\frac{\pi}{6} - x$ (2) $x - \frac{\pi}{6}$ (3) $2x - \frac{\pi}{3}$ (4) $\frac{\pi}{3} - x$
14. Let $x^k + y^k = a^k$ ($a, k > 0$) and $\frac{dy}{dx} + \left(\frac{y}{x} \right)^{\frac{1}{3}} = 0$, then k is : **[JEE (Main) 2020]**
- (1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) $\frac{4}{3}$ (4) $\frac{3}{2}$
15. If $y(\alpha) = \sqrt{2 \left(\frac{\tan \alpha + \cot \alpha}{1 + \tan^2 \alpha} \right) + \frac{1}{\sin^2 \alpha}}$, $\alpha \in \left(\frac{3\pi}{4}, \pi \right)$ then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is : **[JEE (Main) 2020]**
- (1) -4 (2) 4 (3) $\frac{4}{3}$ (4) $\frac{1}{4}$

16. Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to : **[JEE (Main) 2020]**
- (1) $-\frac{\sqrt{5}}{4}$ (2) $\frac{\sqrt{5}}{2}$ (3) $-\frac{\sqrt{5}}{2}$ (4) $\frac{2}{\sqrt{5}}$
17. Let $f(x) = (\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1$, $|x| > 1$. If $\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (\sin^{-1}(f(x)))$ and $y(\sqrt{3}) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to : **[JEE (Main) 2020]**
- (1) $\frac{5\pi}{6}$ (2) $-\frac{\pi}{6}$ (3) $\frac{2\pi}{3}$ (4) $\frac{\pi}{3}$
18. Let f and g be differentiable functions on $\mathbb{R}$ such that $f \circ g$ is the identity function. If for some $a, b \in \mathbb{R}$, $g'(a) = 5$ and $g(a) = b$, then $f'(b)$ is equal to : **[JEE (Main) 2020]**
- (1) 1 (2) $\frac{1}{5}$ (3) 5 (4) $\frac{2}{5}$
19. If $x = 2\sin\theta - \sin 2\theta$ and $y = 2\cos\theta - \cos 2\theta$, $\theta \in [0, 2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta = \pi$ is : **[JEE (Main) 2020]**
- (1) $\frac{3}{2}$ (2) $\frac{3}{4}$ (3) $-\frac{3}{8}$ (4) $-\frac{3}{4}$

EXERCISE # 5 (JA)

SECTION-1

1. For $x > 0$, $\lim_{x \rightarrow 0} ((\sin x)^{1/x} + (1/x)^{\sin x})$ is :- [JEE 2006, 3]
 (A) 0 (B) -1 (C) 1 (D) 2
2. $\frac{d^2x}{dy^2}$ equals :-
 (A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$
 (C) $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$ (D) $-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$
3. (a) Let $g(x) = \ln f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x f(x)$. then for $N = 1, 2, 3, \dots$, $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$
 (A) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$ (B) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$
 (C) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$ (D) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$
- (b) Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$, and $f(x) = g(x) \sin x$.
Statement-1 : $\lim_{x \rightarrow 0} [g(x) \cot x - g(0) \operatorname{cosec} x] = f''(0)$
 and
Statement-2 : $f'(0) = g(0)$
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true. [JEE 2008, 3 + 3]
4. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is [JEE 2009, 4]
5. Let $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. then the value of $\frac{d}{d(\tan \theta)} (f(\theta))$ is [JEE 2011, 4]

6. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point (1,3) is
[JEE(Advanced)-2014, 3]

SECTION-2

7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that
 $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then-
[JEE(Advanced)-2016, 4(-2)]

(A) $g'(2) = \frac{1}{15}$ (B) $h'(1) = 666$ (C) $h(0) = 16$ (D) $h(g(3)) = 36$

8. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as
[JEE(Advanced)-2018, 4(0)]

$$f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1 + (x+j)(x+j-1)} \right) \text{ for all } x \in (0, \infty)$$

(Here, the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$)

(A) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$

(B) $\sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$

(C) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$

(D) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

EXERCISE # 6

MULTIPLE CORRECT

1. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f , then
 (A) $g'(2\pi) = \frac{7}{3}$ (B) $g'(2\pi) = \frac{3}{7}$ (C) $g''(2\pi) = \frac{7}{3}$ (D) $g''(2\pi) = 0$

2. If $f(x) = x \cdot |x|$, then its derivative is :
 (A) $2x$ (B) $-2x$ (C) $2|x|$ (D) $2x \operatorname{sgn} x$

3. If $y = \tan x \tan 2x \tan 3x$, ($\sin 12x \neq 0$) then $\frac{dy}{dx}$ has the value equal to
 (A) $3 \sec^2 3x \tan x \tan 2x + \sec^2 x \tan 2x \tan 3x + 2 \sec^2 2x \tan 3x \tan x$
 (B) $2y (\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$
 (C) $3 \sec^2 3x - 2 \sec^2 2x - \sec^2 x$
 (D) $\sec^2 x + 2 \sec^2 2x + 3 \sec^2 3x$

4. Which of the following statements are true ?
 (A) If $xe^{xy} = y + \sin^2 x$, then at $x = 0$, $(dy/dx) = 1$.
 (B) If $f(x) \equiv a_0 x^{2m+1} + a_1 x^{2m} + a_3 x^{2m-1} + \dots + a_{2m+1} = 0$ ($a_0 \neq 0$) is a polynomial equation with rational coefficients then the equation $f'(x) = 0$ must have a real root. ($m \in \mathbb{N}$).
 (C) If $(x - r)$ is a factor of the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0$ repeated m times where $1 \leq m \leq n$ then r is a root of the equation $f'(x) = 0$ repeated $(m-1)$ times.
 (D) If $y = \sin^{-1}(\cos \sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x)$ then $\frac{dy}{dx}$ is dependent on x .

5. Let $f(x) = \frac{\sqrt{x-2}\sqrt{x-1}}{\sqrt{x-1}-1} \cdot x$ then
 (A) $f'(10) = 1$ (B) $f'(3/2) = -1$
 (C) domain of $f(x)$ is $x \geq 1$ (D) none

6. $\lim_{x \rightarrow \pi} \frac{x^\pi - \pi^x}{x^x - \pi^\pi}$ is equal to –

(A) $\log_{\pi e} \left(\frac{e}{\pi} \right)$

(B) $\log_{\pi e} \left(\frac{\pi}{e} \right)$

(C) $\tan(\cot^{-1}(\ln \pi) - \cot^{-1}(1))$

(D) $\tan(\tan^{-1}(1) - \tan^{-1}(\ln \pi))$

7. Let $P(x)$ be the polynomial $x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$. If $P(-3) = P(2) = 0$ and $P'(-3) < 0$, which of the following is a possible value of 'c' ?

(A) -27

(B) -18

(C) -6

(D) -3

8. Two functions f & g have first & second derivatives at $x = 0$ & satisfy the relations,

$f(0) = \frac{2}{g(0)}, f'(0) = 2g'(0) = 4g(0), g''(0) = 5f''(0) = 6f(0) = 3$ then -

(A) if $h(x) = \frac{f(x)}{g(x)}$ then $h'(0) = \frac{15}{4}$

(B) if $k(x) = f(x) \cdot g(x) \sin x$ then $k'(0) = 2$

(C) $\lim_{x \rightarrow 0} \frac{g'(x)}{f'(x)} = \frac{1}{2}$

(D) none

9. If $y = x^{(\ln x)^{\ln(\ln x)}}$, then $\frac{dy}{dx}$ is equal to :

(A) $\frac{y}{x} (\ln x^{\ln x - 1} + 2 \ln x \ln(\ln x))$

(B) $\frac{y}{x} (\ln x)^{\ln(\ln x)} (2 \ln(\ln x) + 1)$

(C) $\frac{y}{x \ln x} ((\ln x)^2 + 2 \ln(\ln x))$

(D) $\frac{y \ln y}{x \ln x} (2 \ln(\ln x) + 1)$

EXERCISE # 7

1. Find a polynomial function $f(x)$ such that $f(2x) = f'(x) f''(x)$.
2. If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ then $(x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = ky$, then find the value of 'k'.
3. Let $y = x \sin kx$. Find the possible value of k for which the differential equation $\frac{d^2y}{dx^2} + y = 2k \cos kx$ holds true for all $x \in \mathbb{R}$.
4. Prove that if $|a_1 \sin x + a_2 \sin 2x + \dots + a_n \sin nx| \leq |\sin x|$ for $x \in \mathbb{R}$, then $|a_1 + 2a_2 + 3a_3 + \dots + na_n| \leq 1$
5. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x^2) \cdot f''(x) = f'(x) \cdot f'(x^2)$ for all real x . Given that $f(1) = 1$ and $f'''(1) = 8$, compute the value of $f'(1) + f''(1)$.
6. Show that $R = \frac{[1 + (dy/dx)^2]^{\frac{3}{2}}}{d^2y/dx^2}$ can be reduced to the form $R^{\frac{2}{3}} = \frac{1}{(d^2y/dx^2)^{\frac{2}{3}}} + \frac{1}{(d^2x/dy^2)^{\frac{2}{3}}}$.
7. Let $f(x) = \frac{\sin x}{x}$ if $x \neq 0$ and $f(0) = 1$. Define the function $f'(x)$ for all x and find $f''(0)$ if it exist.
8. Suppose f and g are two functions such that $f, g: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \ln(1 + \sqrt{1+x^2})$ and $g(x) = \ln(x + \sqrt{1+x^2})$ then find the value of $x e^{g(x)} \left(f\left(\frac{1}{x}\right) \right)' + g'(x)$ at $x = 1$.
9. Let $\frac{f(x+y) - f(x)}{2} = \frac{f(y) - a}{2} + xy$ for all real x and y . If $f(x)$ is differentiable and $f'(0)$ exists for all real permissible values of 'a' and is equal to $\sqrt{5a-1-a^2}$. Prove that $f(x)$ is positive for all real x .
10. If $y = \tan^{-1} \frac{x}{1+\sqrt{1-x^2}} + \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$, then find $\frac{dy}{dx}$ for $x \in (-1, 1)$.
11. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for all $x \in \mathbb{R}$, then prove that $f(2) = f(1) - f(0)$.
12. If α be a repeated root of a quadratic equation $f(x) = 0$ & $A(x)$, $B(x)$, $C(x)$ be the polynomials of degree 3, 4 & 5 respectively, then show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by $f(x)$, where dash denotes the derivative

ANSWER KEY

EXERCISE # 1

1.	A	2.	B	3.	B	4.	D	5.	B	6.	C	7.	B
8.	B	9.	C	10.	C	11.	D	12.	A	13.	A	14.	C
15.	D	16.	A	17.	A	18.	B	19.	A	20.	B	21.	C
22.	B	23.	C	24.	B	25.	B	26.	C	27.	D	28.	C
29.	D	30.	B	31.	A	32.	A	33.	A	34.	D	35.	A
36.	D	37.	C	38.	D	39.	B	40.	C	41.	A	42.	D
43.	A	44.	D	45.	C								

EXERCISE # 2

1. 16

2. (a) $Dy = (\cos x)^{\ln x} \left[\frac{\ln(\cos x)}{x} - \tan x \ln x \right] + (\ln x)^x \left[\frac{1}{\ln x} + \ln(\ln x) \right];$

(b) $\frac{dy}{dx} = e^{x^{e^x}} \cdot x^{e^x} \left[\frac{e^x}{x} + e^x \ln x \right] + e^{x^{e^x}} x^{e^x-1} x^{e^x} [1 + e \ln x] + x^{e^{e^x}} e^{e^x} \left[\frac{1}{x} + e^x \ln x \right]$

4. $\frac{y}{x} \cdot \frac{x \ln x + x \ln x \cdot \ln y + 1}{\ln x (1 - x - y \ln a)}$

7.

8

9.

$\frac{1 + \sqrt{1 - x^4}}{x^6}$

10. $f(x) = \begin{cases} -\frac{2}{3} \left[\frac{1}{6} + \ln \frac{3}{2} \right] x & \text{if } x \leq 0 \\ \left(\frac{1+x}{2+x} \right)^{1/x} & \text{if } x > 0 \end{cases}$

13.

100

14.

$\frac{32}{16 + \pi^2} - \frac{8}{\ln 2}$

15. (a) $\left(-\frac{1}{2}, \frac{1}{2} \right), (-\infty, \infty);$ (b) $f(x) = \frac{2x}{\sqrt{1-4x^2}};$ (c) $\frac{16\sqrt{3}}{9}$

16. (a) $1/6;$ (b) $\frac{1}{14};$ (c) -1 17. $\frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$

19. $\frac{1}{2}$ or $-\frac{1}{2}$ 20. 3 21. (a)(i) $y'(0) = -1;$ (ii) $y''(0) = 2;$ (b) $a = 1, -2$

22. $-\frac{3}{2}$ 23. $-\frac{9}{4}$ 24. $2(1+2x) \cdot \cos 2(x+x^2)$ 25. 100

EXERCISE # 3

1. $\frac{5}{6}$ 2. $\frac{1}{6}$ 3. $-\frac{1}{3}$ 4. $-\frac{1}{2}$ 5. 1000 6. 11
7. f is cont. but not derivable at $x = 0$ 8. (a) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n}$; (b) a_1 ; (c) a_n
10. (A) S; (B) P; (C) R

EXERCISE # 4 (JM)

1. 4 2. 3 3. 2 4. 2 5. 1 6. 4 7. 3
8. 2 9. 2 10. 4 11. 2 12. 1 13. 2 14. 1
15. 2 16. 3 17. 2 [There will be ans. Correction in NTA answer key]
18. 2 19. [Bonus]

EXERCISE # 5 (JA)

SECTION-1

1. C 2. D 3. (a) A; (b) A 4. 2 5. 1
6. 8

SECTION-2

7. B,C 8. D

EXERCISE # 6

1. BD 2. CD 3. ABC 4. AC 5. AB 6. ACD 7. A
8. ABC 9. BD

EXERCISE # 7

1. $\frac{4x^3}{9}$ 2. 25 3. $k = 1, -1$ or 0 5. 6
7. $f'(x) = \begin{cases} \frac{x \cos x - \sin x}{x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$; $f''(0) = -\frac{1}{3}$ 8. Zero 10. $\frac{1-2x}{2\sqrt{1-x^2}}$

Result JEE Main

Result Highlights 2019



100 Percentile
HIMANSHU GAURAV SINGH
2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



99.98 Percentile
SARTHAK ROUT
2020 (CCP)



99.98 Percentile
VIBHAV AGGARWAL
2019 (CCP)



99.97 Percentile
RITVIK GUPTA
2020 (DLP)



99.97 Percentile
BHAVYA JAIN
2020 (CCP)



99.96 Percentile
AYUSH PATTNAIK
2019 (CCP)



99.96 Percentile
SAYANTAN DHAR
2020 (DLP)

Result Highlights 2018

AIR
29



Sarthak Behera

AIR
36



Shashank Agrawal

AIR
46



Shreya Pathak

AIR
48



Raaghav Raaj

AIR
2
SC



Aayush Kadam

AIR
18
OBC



Gunjan

Result Highlights 2017

AIR
12



Abhay Goyal

AIR
14



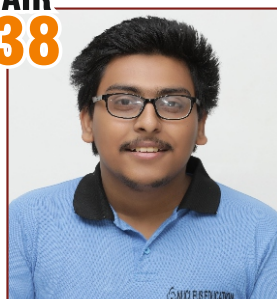
Piyush Tibarwal

AIR
20



Abhinav Kumar

AIR
38



Tushar Gautam

AIR
49



Saurav Sahoo

AIR
60



Pratyush

Mentored by Team

AIR
4



Kushal Babel
2014

AIR
15



Gourika Bang
2015

AIR
21



Amit Malav
2013

AIR
24



Suraj Sanjay Jog
2012

AIR
38



Chitraang Murdia
2014

AIR
58



Pulkit Tandon
2012

AIR
68



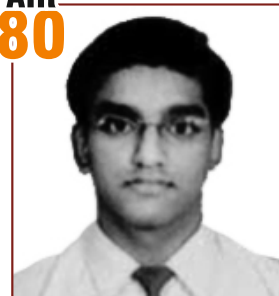
Kapil Shobhnani
2016

AIR
70



Utkarsh Sinha
2015

AIR
80



S John S
2015